

Prognostic factors for surgical treatment of radiation-induced scleral necrosis after brachytherapy for uveal melanoma

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Abstract

Objectives:

Radiation-induced scleral necrosis (RISN) is a rare complication of brachytherapy for uveal melanoma, and may require surgical treatment in selected cases. We aimed to identify the prognostic factors for RISN treatment.

Design:

Retrospective, single-center, observational case series

Methods:

All patients with brachytherapy for uveal melanoma treated at our institution between 01/1999 and 12/2016 who developed RISN were followed until 02/2021. Various parameters were evaluated through univariable and multivariable Cox regression analysis. The surgical intervention due to RISN was the principal outcome event of this study.

Results:

Of 115 patients in the final cohort, 51 individuals (44.3%) underwent RISN treatment (conjunctival revision [$n = 2$], patching [$n = 46$] or enucleation [$n = 3$]) at median 1.80 months after RISN occurrence. Significant RISN characteristics were summarized into a novel RISN severity scale – Grade I: largest diameter ≤ 5 mm and no progression; Grade II: largest diameter > 5 mm or any progression during the follow-up; Grade III: presence of uveal prolapse; and Grade IV: leakage through open eyewall perforation. In the multivariable analysis, the RISN severity scale (aHR = 2.37 per grade increase, $p = 0.005$) and the time between brachytherapy and RISN occurrence (< 15 months, aHR = 6.33, $p < 0.0001$) were independently associated with the study endpoint. The RISN severity scale showed high diagnostic accuracy for prediction of RISN treatment (AUC = 0.869).

Conclusions:

In our series, about the half of RISN cases underwent surgical treatment. The presented novel severity scale for RISN might become a helpful tool for clinical management of individuals with RISN. We recommend external validation of the diagnostic accuracy of the presented scale.

Introduction

Since brachytherapy has been established as standard therapy for small- and middle-sized uveal melanoma (UM) ¹ the management of its possible complications has great importance for therapy success. The radiation induced complications can be divided into early ones, that can occur within first weeks after brachytherapy (exudative retinal detachment, choroidal detachment, extensive tumor necrosis with toxic tumor syndrome, and conjunctival dehiscence) and delayed ones occurring up to years after the radiation (neovascular glaucoma, cataract, radiation retinopathy and radiation-induced optic neuropathy, vitreous hemorrhage) ²⁻⁵. Timely identification and treatment of these complications is essential for preservation of the visual function and the eye itself.

Radiation-induced scleral necrosis (RISN) is considered as a rare complication after brachytherapy of UM with the incidence of about 1–11% in large series ^{2, 5-9}. Depending on the severity and clinical course, RISN might require a special treatment in selected cases ^{5, 10-12}. While for stable and mild RISN cases proper management consists of long-term observation, progressive RISN can threaten the tectonic stability of the eye and require surgical treatment like patch grafting ^{5, 12}. Furthermore, a RISN complicated with a leaky wound of an eyewall perforation might not be eligible for reconstructive surgery and therefore necessitate enucleation of the affected eye ^{5, 10, 12}.

As RISN is a rare complication, the literature on its management and, particularly, the evidence on the prognostic factors requiring RISN treatment are sparse. To the best of the authors' knowledge, there is no validated classification of RISN severity. In order to fill this gap, we analyzed the course of RISN after brachytherapy for UM to identify the prognostic factors for RISN treatment. Based on the significant RISN characteristics, we have established a novel severity scale for RISN.

Materials And Methods

Study design and study population

This retrospective study was approved by the Institutional Ethics Committee (Ethik-Kommission, Medizinische Fakultät der Universität Duisburg-Essen, registration number 18-8190-BO) and registered in the German clinical trial registry (DRKS, Unique identifier: DRKS00019049). The study is based on the institutional observational cohort with 4689 UM patients who underwent brachytherapy at the Clinic for Ophthalmology and Radiology of University Hospital of Essen between January 1st, 1999 and December 31st, 2016. All cases with RISN documented until August 31st, 2019 were included in the final analysis.

Detailed description of UM management was published previously ⁶. In short, the tumors with largest initial tumor thickness up to 6.5 mm were treated with ¹⁰⁶Ru plaques. For larger tumors, bi-nuclide ¹⁰⁶Ru-¹²⁵I plaques were used. The common prescribed tumor apex dose was ≥ 130 Gy for ruthenium or ranged between 70–100 Gy for bi-nuclide plaques respectively. The common prescribed sclera dose was 700–1500 Gy.

For tumor control and surveillance of radiation-related complications after brachytherapy, all patients underwent regular follow-ups at our outpatient service. According to the institutional policy, the initial follow-ups were performed in 3-months intervals in the first year. In cases with stable disease, the intervals were prolonged up to once every year. After first RISN detection patients with RISN underwent follow-ups at shorter time intervals (4–6 weeks), depending on the severity of necrotic changes in the sclera.

There was an individual assessment of the indication to RISN treatment based on the initial characteristics and further clinical course of RISN. In particular, a full thickness scleral perforation and conjunctival dehiscence persistent longer than 4 weeks represented an absolute indication for surgical intervention on RISN. In addition, the documented progression of RISN during follow-up and scleral thinning resulting in uveal prolapse were accounted in favor of surgical treatment of RISN, also taking into consideration the patients' clinical condition, age and preferences. RISN was treated using different biological graft materials for patching (e.g. scleral, corneal patches), whereas enucleation of the eye was performed in cases of non-treatable eyewall perforation and irreparable RISN.

Data management

The electronic patient records were reviewed for initial and follow-up data until February 28th, 2021. The variables of the interests collected from the institutional UM database included the general (patient, tumor and treatment) and RISN-related characteristics.

The following general variables were recorded: demographics (age and sex), medical history (arterial hypertension, diabetes mellitus), anatomical tumor location (choroid, ciliary body (CB), iris), anterior tumor margin (iris, CB, anterior to equator, posterior to equator), posterior tumor margin (posterior to equator, anterior to equator, CB), quadrant tumor location (superior, superior-nasal, nasal, inferior-nasal, inferior, inferior-temporal, temporal, superior-temporal), largest tumor thickness and largest basal tumor diameter (LBSD) prior to the therapy (based on ultrasonography), extraocular tumor extension (EOE), plaque type (bi-nuclide $^{106}\text{Ru}/^{125}\text{I}$ vs. ^{106}Ru), apex and sclera dose, extraocular muscles disinsertion during plaque insertion, mean and maximal intraocular pressure based on all records routinely measured at each patient visit after brachytherapy (mm of mercury), development of metastasis, the indication to enucleation (incl. the underlying cause), different time intervals (between brachytherapy, RISN detection/treatment, and the last follow-up).

Patients with a pigmentation of the outer scleral surface at the tumor side at first clinical examination were considered as cases with EOE. All twelve patients in the cohort had a circumscribed area of EOE. A presence of persistent disruption of conjunctival integrity at the irradiated side without re-epithelialization for over four weeks was regarded as conjunctival dehiscence and recorded for further analysis.

The variables regarding the RISN characteristics were evaluated using slit lamp examination and photodocumentation, including largest dimension (LD) of RISN at first diagnosis and at last follow-up or

pre-treatment, LD/LBTD-ratio, presence of uveal prolapse or herniation through the necrotic sclera, presence of eyewall perforation within necrotic sclera and leakage.

Study endpoints and statistical analysis

The endpoint of the study was the identification of the parameters associated with the indication to RISN treatment, whereat the significant RISN-related characteristics were summarized into a novel RISN severity scale. Accordingly, surgical intervention of any type (conjunctival revision surgery with conjunctival plastic reconstruction, scleral patch grafting or enucleation) due to RISN was the principal outcome event of this study. Data analysis was performed with the use of SPSS (version 25, SPSS Inc., IBM, Chicago, IL, USA). Differences with a P-value of 0.05 or less were considered as significant. The data were reported as median values (with interquartile range [IQR] between the 25. and 75. percentile) or absolute numbers (with percentages) for continuous and nominal variables respectively.

Associations between the potential predictors and the treatment necessity were evaluated using the univariable and multivariable Cox regression analysis. Continuous variables with the differences in the univariable analysis at a P-value of 0.1 or less were additionally assessed as dichotomous variables using the cutoffs upon the receiver operating characteristic (ROC) curves. The diagnostic accuracy of the severity scale and the final prediction model were assessed upon the area under the curve (AUC) in the ROC analysis.

Results

Patient population

Between 01/1999 and 08/2019, 115 UM patients developed RISN after brachytherapy and therefore were included in this study. The median age of patients was 62.4 years (IQR: 48.6–70.2) and 68 (59.1%) of them were female. The median post-treatment follow-up was 55.0 months (IQR: 29.5–88.6). RISN was documented at the median time interval of 18.5 months (IQR: 9.8–33.8) after brachytherapy. The detailed information on the characteristics of the analyzed cohort is listed in Table 1.

Table 1
Major baseline and follow-up characteristics of the cohort

Parameter	Number of cases (%) or median value (IQR)
General characteristics	
Age	62.4 (48.6–70.2)
Sex (Female)	68 (59.1%)
Largest tumor thickness prior to the therapy, mm	6.5 (4.4 -9.0)
LBTD, mm	13.1 (10.6–16.1)
Reduction of tumor thickness after brachytherapy, %	62.2 (35.5–77.9)
Ciliary body involvement	98 (85.2%)
Iris involvement	4 (3.5%)
Anterior tumor margin*:	16 (13.9%)
Anterior to equator	94 (81.7%)
Ciliary body	4 (3.5%)
Iris	
Posterior tumor margin**:	48 (41.7%)
Anterior to equator	50 (43.5%)
Posterior to equator	14 (12.2%)
Ciliary body	

Abbreviations: IQR – interquartile range; RISN- radiation induced scleral necrosis, LD- largest dimension, LBTD-Largest basal tumor diameter; CI- Confidence interval; *-No available information by 1 patient; **- No available information by 3 patients

Parameter	Number of cases (%) or median value (IQR)
Meridional tumor location	10 (8.7%)
Nasal	16 (13.9%)
Nasal superior	11 (9.6%)
Nasal inferior	17 (14.8%)
Temporal	12 (10.4%)
Temporal superior	9 (7.8%)
Temporal inferior	11 (9.6%)
Superior	12 (10.4%)
Inferior	
Extraocular extension	12 (10.4%)
Muscle disinsertion:	26 (22.6%)
Inferior rectus muscle	27 (23.5%)
Superior rectus muscle	24 (20.9%)
Lateral rectus muscle	21 (18.3%)
Medial rectus muscle	14 (12.2%)
Superior oblique muscle	12 (10.4%)
Inferior oblique muscle	
Plaque type	46 (40%)
bi-nuclide (^{106}Ru - ^{125}I)	69 (60%)
^{106}Ru plaque	
Radiation duration, hour	145.6 (101.9-191.7)
Scleral contact dose, Gy	1032.0 (762.0-1344.0)
Dose rate, Gy/h	6.8 (5.9–8.2)
Conjunctival dehiscence	7 (6%)
Glaucoma medication	27 (23.5%)

Abbreviations: IQR – interquartile range; RISN- radiation induced scleral necrosis, LD- largest dimension, LBTD-Largest basal tumor diameter; CI- Confidence interval; *-No available information by 1 patient; **- No available information by 3 patients

Parameter	Number of cases (%) or median value (IQR)
Maximal intraocular pressure after brachytherapy, mmHg	20.0 (17.0–24.0)
Mean intraocular pressure after brachytherapy, mmHg	15.0 (13.0–17.7)
Diabetes mellitus	6 (5.2%)
Arterial hypertension	46 (40.0%)
Time intervals, months:	55.0 (29.5–88.6)
Brachytherapy > last follow-up	18.5 (9.8–33.8)
Brachytherapy > RISN diagnosis	1.8 (0.7–8.3)
RISN diagnosis > RISN-Therapy	
RISN-related characteristics	
LD of RISN at diagnosis, mm	4.20 (2.20–6.00)
LD of RISN at last follow-up or pre-treatment, mm	4.90 (3.00–7.00)
Ratio: LD at diagnosis / LBTD	27.78 (17.24–48.63)
Ratio: LD at last follow-up or pre-treatment / LBTD	40.02 (20.20–55.00)
RISN course	3 (2.6%)
Decrease of LD	30 (26%)
Increase of LD	5 (4.3%)
without therapy	25 (21.7%)
with therapy	33 (28.7%)
Uveal prolapse	6 (5.2%)
Globe perforation	
Abbreviations: IQR – interquartile range; RISN- radiation induced scleral necrosis, LD- largest dimension, LBTD-Largest basal tumor diameter; CI- Confidence interval; *-No available information by 1 patient; **- No available information by 3 patients	

RISN: characteristics and the clinical course

The median LD of RISN was 4.2 mm (IQR: 2.2–6.0) at first diagnosis, and 4.9 mm (IQR: 3.0–7.0) at the last follow-up or pre-treatment, whereat 30 individuals showed an increase and three ones a decrease of the LD of RISN during the observation time (Fig. 1). Globe perforation was diagnosed in six cases.

The median follow-up after RISN diagnosis in the sub-cohort without RISN treatment ($n = 64$, 55.7%) was 27.2 months (IQR: 6.1–50.6). Surgical treatment of RISN was performed in 51 patients (44.3%) at the time interval between two days and 207.5 months (median: 1.80 months; IQR: 0.7–8.4) after first diagnosis. Of 48 cases with globe sparing surgical intervention on RISN, only 2 treatments were limited to conjunctival revision surgery with conjunctival plastic reconstruction and 46 patients underwent scleral patch grafting (with conjunctival revision). Conjunctival revision surgery with conjunctival plastic reconstruction was performed at median 1.5 months (IQR: 0.6–2.4) after RISN diagnosis. RISN treatment using scleral patch grafting was initiated at median time interval of 1.9 months (IQR: 0.9–8.6). Three patients with full-thickness scleral defect due to RISN and with extrusion of vitreous body underwent enucleation after 1.2 months (IQR: 0.1–9.4).

Moreover, nine other patients in the treated RISN sub-cohort and six individuals in the untreated RISN sub-cohort underwent eye enucleation due to the following reasons: suspected tumor recurrence ($n = 10$), secondary glaucoma with painful eye ($n = 1$), persistent retinal detachment with blindness ($n = 1$), phthisis bulbi ($n = 2$), a blind eye with progressive severe RISN after scleral patching ($n = 1$). Therefore, patients with treated RISN were at higher risk for enucleation during the whole post-brachytherapy follow-up period (hazard ratio [HR] = 2.19, 95% confidence interval [CI] = 1.14–4.42, $p = 0.019$) regardless the indication to enucleation. In addition, individuals with treated RISN were at higher risk for distant metastases in the further course of disease (HR = 2.53, 95% CI = 1.35–4.73, $p = 0.002$).

RISN severity scale

The following RISN-related characteristics showed significant association with the indication to RISN treatment: LD > 5 mm at the last follow-up ($p = 0.021$), increase of the RISN size ($p = 0.002$), herniation or protrusion of underlying tissue through the necrotic sclera ($p < 0.0001$) and globe perforation (< 0.0001 , table 2). Based on these parameters, we have established a severity scale for RISN (grades I-IV, Fig. 2). Accordingly, the cases without any of the above-mentioned prognostic factors were regarded as the Grade I. A RISN with documented progression or with the LD > 5mm without further risk factors was referred to the Grade II. In turn, the Grade III RISN was characterized by the scleral thinning and herniation/prolapse of underlying tissue (uvea/necrotic tumor mass). Finally, Grade IV RISN presented with a full-thickness scleral perforation, globe hypotony and extrusion of the ocular content. The change in the RISN severity during the whole documented observational period is shown in the appropriate Sankey diagram (Fig. 3).

The Cox regression analysis showed strong correlation between the severity scale and RISN treatment ($p < 0.0001$). In addition, the ROC analysis confirmed sufficient diagnostic accuracy of the severity scale for the prediction of RISN treatment (AUC = 0.869, see supplementary Figure E1). Kaplan-Meier survival plot visualized the differences in the clinical course of RISN depending on the severity grade (Fig. 4). The higher the grade of RISN severity, the rather and earlier underwent the patients the RISN treatment.

When assessing the extent of RISN treatment as a categorical variable with stepwise increasing treatment invasiveness (conjunctival plastic reconstruction vs. patch grafting vs. enucleation), the RISN

severity scale also showed strong correlation with the treatment extent ($p < 0.0001$, $r = 0.712$; see also Table 3 with the frequency of different RISN management options depending on the RISN severity grade). Finally, there was a significant association of the RISN severity scale with the risk of enucleation at any timepoint and regardless the indication ($p = 0.0284$).

Table 2

Univariable Cox regression analysis for the association between different (general and RISN-related) characteristics and RISN treatment

Parameter	HR (95% CI)	p-value
Time interval brachytherapy > RISN diagnosis	0.98 (0.96–0.99)	0.010
Time interval brachytherapy > RISN diagnosis under 15months	3.75 (2.06–6.83)	<0.0001
Age	1.00 (0.98–1.02)	0.909
Sex	1.25 (0.70–2.23)	0.444
Scleral contact dose	1.00 (1.00–1.00)	0.405
Dose rate, per Gy/h	1.08 (0.95–1.22)	0.267
Radiation duration, per hour	1.00 (0.99–1.01)	0.336
Plaque type: bi-nuclide (^{106}Ru - + ^{125}I) vs. ^{106}Ru	1.37 (0.78–2.41)	0.269
Ciliary body involvement	0.84 (0.38–1.87)	0.667
Iris involvement	0.89 (0.22–3.65)	0.866
Extraocular extension	2.53 (1.22–5.26)	0.013
Posterior tumor margin (posterior to equator vs anterior to equator)	0.87 (0.49–1.56)	0.645
Anterior tumor margin	1.03 (0.44–2.43)	0.944
Meridional tumor location: nasal	1.36 (0.61–3.02)	0.456
Meridional Tumor Location: nasal superior	0.57 (0.24–1.34)	0.196
Meridional Tumor Location: superior	0.97 (0.38–2.44)	0.946
Meridional Tumor Location: temporal superior	1.04 (0.44–2.45)	0.926
Meridional Tumor Location: temporal	1.55 (0.79–3.03)	0.199
Meridional Tumor Location: temporal inferior	1.31 (0.56–3.08)	0.534
Meridional Tumor Location: inferior	1.14 (0.49–2.68)	0.764
Meridional Tumor Location: nasal inferior	0.50 (0.16–1.62)	0.250
Conjunctival dehiscence	2.35 (1.00-5.54)	0.050
Largest tumor thickness prior to the therapy, mm	1.12 (1.01–1.25)	0.041
Largest tumor thickness prior to the therapy, over 7.8mm	2.00 (1.14–3.51)	0.016
Reduction of tumor thickness after brachytherapy, %	1.00 (0.99–1.01)	0.576
Abbreviations: LBTD-Largest basal tumor diameter, LD-largest diameter, RISN- radiation-induced scleral necrosis; CI- Confidence interval; HR- hazard ratio		

Parameter	HR (95% CI)	p-value
LBTD mm	1.09 (0.99–1.20)	0.098
LBTD over 14mm	2.34 (1.21–4.54)	0.011
Inferior rectus muscle	0.71 (0.35–1.47)	0.358
Superior rectus muscle	0.79 (0.39–1.63)	0.521
Lateral rectus muscle	1.71 (0.90–3.22)	0.099
Medial rectus muscle	0.96 (0.48–1.92)	0.910
Superior oblique muscle	1.92 (0.90–4.10)	0.094
Inferior oblique muscle	0.38 (0.09–1.56)	0.177
Maximal intraocular pressure after brachytherapy mm HG	1.02 (0.98–1.05)	0.414
Mean intraocular pressure after brachytherapy mm HG	1.05 (0.98–1.11)	0.176
Glaucoma medication	1.17 (0.62–2.20)	0.635
Diabetes mellitus	1.04 (0.32–3.34)	0.952
Arterial hypertension	0.68 (0.38–1.22)	0.196
Ratio: RISN LD at diagnosis / LBTD (%)	1.00 (0.98–1.02)	0.961
Ratio: RISN LD at last follow-up or pre-treatment / LBTD (%)	1.01 (1.00-1.02)	0.199
LD of RISN at diagnosis (mm)	1.01 (0.90–1.13)	0.928
LD of RISN at last follow-up or pre-treatment(mm)	1.13 (1.00-1.27)	0.045
LD of RISN at last follow-up or pre-treatment (mm) over 5mm	2.26 (1.13–4.51)	0.021
RISN size increase in follow up	2.51 (1.42–4.42)	0.002
RISN with uveal prolapse	3.36 (1.90–5.93)	< 0.0001
RISN with globe perforation	7.36 (3.06–17.73)	< 0.0001
Severity scale	3.50 (2.47–4.95)	< 0.0001
Abbreviations: LBTD-Largest basal tumor diameter, LD-largest diameter, RISN- radiation-induced scleral necrosis; CI- Confidence interval; HR- hazard ratio		

Table 3

The rate of different RISN management modalities depending on the RISN severity grade

RISN severity	RISN management			
	<i>observation</i>	<i>conjunctival plastic reconstruction</i>	<i>scleral patch grafting</i>	<i>enucleation</i>
Grade I	45	2	0	0
Grade II	11	0	18	0
Grade III	8	0	25	0
Grade IV	0	0	3	3
Abbreviations: RISN- radiation-induced scleral necrosis.				

Predictive value of patient and tumor characteristics

Shorter time interval between brachytherapy and RISN occurrence (< 15 months, $p < 0.0001$), presence of conjunctival dehiscence ($p = 0.05$) and different tumor features like EOE ($p = 0.013$), maximal tumor thickness (> 7.8 mm, $p = 0.016$), and LBTD (> 14 mm, $p = 0.011$) showed significant associations with the RISN treatment in the univariable Cox regression analysis (Table 2). In the multivariable Cox regression analysis (Table 4), earlier occurrence of RISN after brachytherapy (< 15 months, adjusted HR [aHR] = 6.33, 95% CI = 2.59–15.49, $p < 0.0001$) and the severity scale (aHR = 2.37 per grade increase, 95% CI = 1.30–4.30, $p = 0.005$) were confirmed as independent predictors of RISN requiring surgical treatment. The prediction model including both parameters (severity scale and RISN timing) showed a higher diagnostic accuracy (AUC = 0.885) for the prediction of treatment necessity than the severity scale alone (supplementary Figure E1).

Table 4
Final multivariable Cox regression analysis of the predictors of RISN treatment

Parameter	aHR (95% CI)	p-value
Time interval brachytherapy to RISN diagnosis, < 15months	6.33 (2.59–15.49)	< 0.0001
Extraocular extension	1.68 (0.76–3.70)	0.200
Conjunctival dehiscence	0.75 (0.25–2.21)	0.599
Largest tumor thickness, > 7.8mm	0.95 (0.36–2.45)	0.908
LBTD, > 14mm	1.48 (0.64–3.44)	0.364
Severity scale, per-grade-increase	2.37 (1.30–4.30)	0.005
Abbreviations: RISN- radiation-induced scleral necrosis; CI- Confidence interval; aHR- adjusted hazard ratio		

Discussion

In this study, we analyzed the course of RISN after brachytherapy for UM. Approximately 50% of patients in our institutional cohort required surgical treatment of RISN. We developed a novel classification scale to grade the severity of the radiation induced scleral damage. This severity scale, as well as the subsequent prediction model consisting of the severity scale and time interval between brachytherapy and RISN diagnosis, reliably predicts the indication to RISN treatment.

Classification of RISN severity

Due to a relatively low incidence, the evidence on the clinical course of RISN is sparse. To analyze the late effects of irradiation on the visual system, Gordon et al presented 1995 a classification scheme for radiation induced damage of different ocular structures¹³. However, arbitrary selection of the classification components and the discrimination between the grades based on subjective assessment necessitates the clarification of the diagnostic accuracy and interobserver reliability of this classification. To avoid these limitations, we performed statistical evaluation of the potential scale components prior to their inclusion in the severity scale. We analyzed and confirmed a high diagnostic accuracy of this novel scale. Finally, the components of the presented novel severity scale are based on clearly defined morphologic changes in the sclera with simply measurable parameters allowing a robust allocation of the scleral damage to the appropriate severity grade.

RISN treatment: rate and prognostic factors

There are just few reports about RISN cases requiring treatment. In the series based on the 5057 cases treated with brachytherapy, Kaliki et al. reported 14% (10/73) and 5% (4/73) of RISN patients treated with scleral patch graft and with enucleation respectively⁵. In contrast, 11 of 15 RISN cases (73%) required surgical therapy with scleral patch and/or conjunctival flap in the UM cohort threatened with ¹⁹⁸Au

brachytherapy¹⁰. In a large institutional series with 301 consecutive patients treated with ¹²⁵I brachytherapy, 6 (2%) patients were enucleated due to RISN¹⁴. In our study, about the half of 115 patients underwent surgery due to RISN, whereas the occurrence of RISN in first 15 months after brachytherapy was significantly associated with the therapy indication. This association between the timing of RISN onset and later therapy requirement might be related to the severity of initial scleral damage after brachytherapy. In particular, the more advanced are radiation-induced changes in the sclera, the earlier and prominent is the breakdown of local repair mechanisms. This results in higher probability of earlier onset of and rapid progression of RISN necessitating surgical treatment.

The essential role of local repair mechanisms for sclera regeneration has already been discussed with regard to the scleral recovery after diathermy¹⁵. Generally, sclera is a dynamic tissue with continuous synthesis and degradation of extracellular matrix¹⁶. Radiation induced scleral alteration can cause an imbalance between these two processes in favor of the degradation. This can cause a progressive scleral necrosis with the protrusion of inactive tumor mass up to full thickness globe perforation. In this context, the possibility of a stable course, or even of a spontaneous partial regression of RISN underlines the importance of the balance between reparative and destructive processes in the sclera after radiation therapy. In particular, we have identified three cases with partially regressive RISN in our cohort. Radin et al had also reported three RISN cases with partial regression in their study.¹¹ Regressive course could be explained by the prevalence of local repair mechanisms like transformation of fibrocytes into active fibroblasts following any insult to the sclera, which are able to synthesize all of the component molecules of the extracellular scleral matrix¹⁷.

Finally, development of RISN requiring treatment was significantly associated with the risk of enucleation (due to any reasons not limited to RISN) and distant metastases in the later course of disease. We presume that the progressive character of RISN could be related to increased destructive radiation burden to the sclera, especially in larger tumors, probably hindering the physiological and reparative processes in the affected radiation site after brachytherapy. Fittingly, different tumor-related characteristics (like EOE and tumor size) were associated with the indication to RISN treatment in our univariable analysis. Of note, the analyzed patient cohort is too small for any causal conclusions and further investigations are needed to explore this correlation.

Limitations

The major limitation of the study is related to the retrospective nature of the presented analysis associated with lower accuracy and incompleteness of the recoded data. The study endpoint, the development of RISN necessitating surgery partially reflects the indications threshold for RISN treatment of the ophthalmologists in our clinic. Therefore, an external validation of the clinical utility of the presented results is mandatory prior to inclusion of the proposed severity scale in the clinical routine.

Conclusion

Depending on the course of reparative and destructive processes in the sclera after brachytherapy for UM, RISN might require surgical treatment even decades after the first diagnosis. We summarized the clinically relevant characteristics of the damaged sclera into a novel RISN severity scale, which could reliably predict the indication to treatment. Along with the RISN severity scale, the time interval between brachytherapy and RISN diagnosis independently predicted the RISN treatment. After successful external validation, the presented RISN severity scale might become a useful tool in clinical management of individuals with RISN after brachytherapy for UM.

Declarations

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3. CONFLICTS OF INTEREST: None.

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Figures

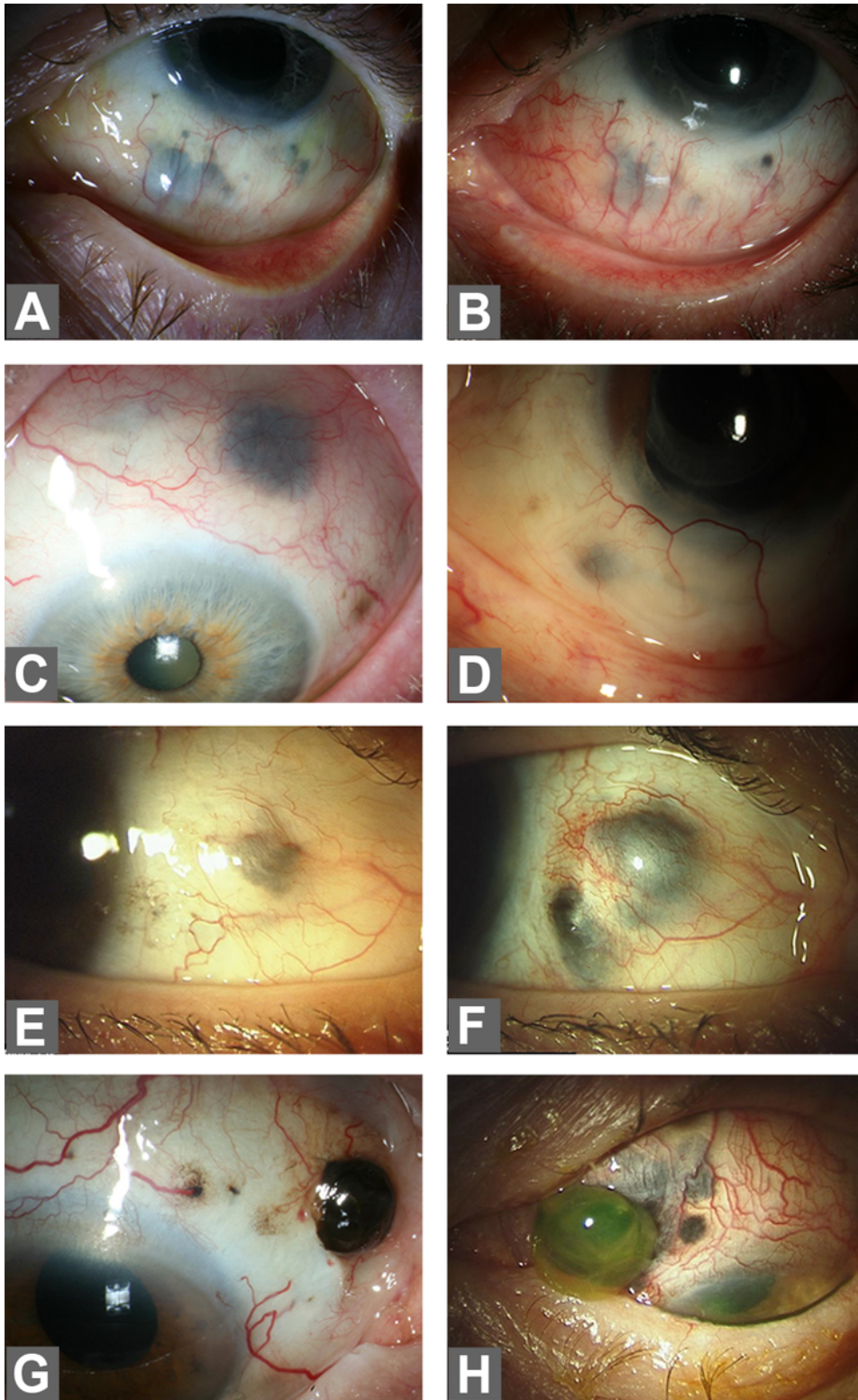


Figure 1

Examples of RISN with different clinical course after brachytherapy for UM: a case with RISN, which developed 36 months after brachytherapy (A) and showed regressive course over 4 years of follow-up (B); two cases (C and D) with RISN Grade I; a case with RISN, which developed 14 months after brachytherapy (E) with progression within 7 months after diagnosis (F); two examples of severe RISN presenting with uveal prolapse (Grade III, G) or globe perforation with extrusion of vitreous body (Grade IV, H).

Abbreviations: RISN- radiation induced scleral necrosis;

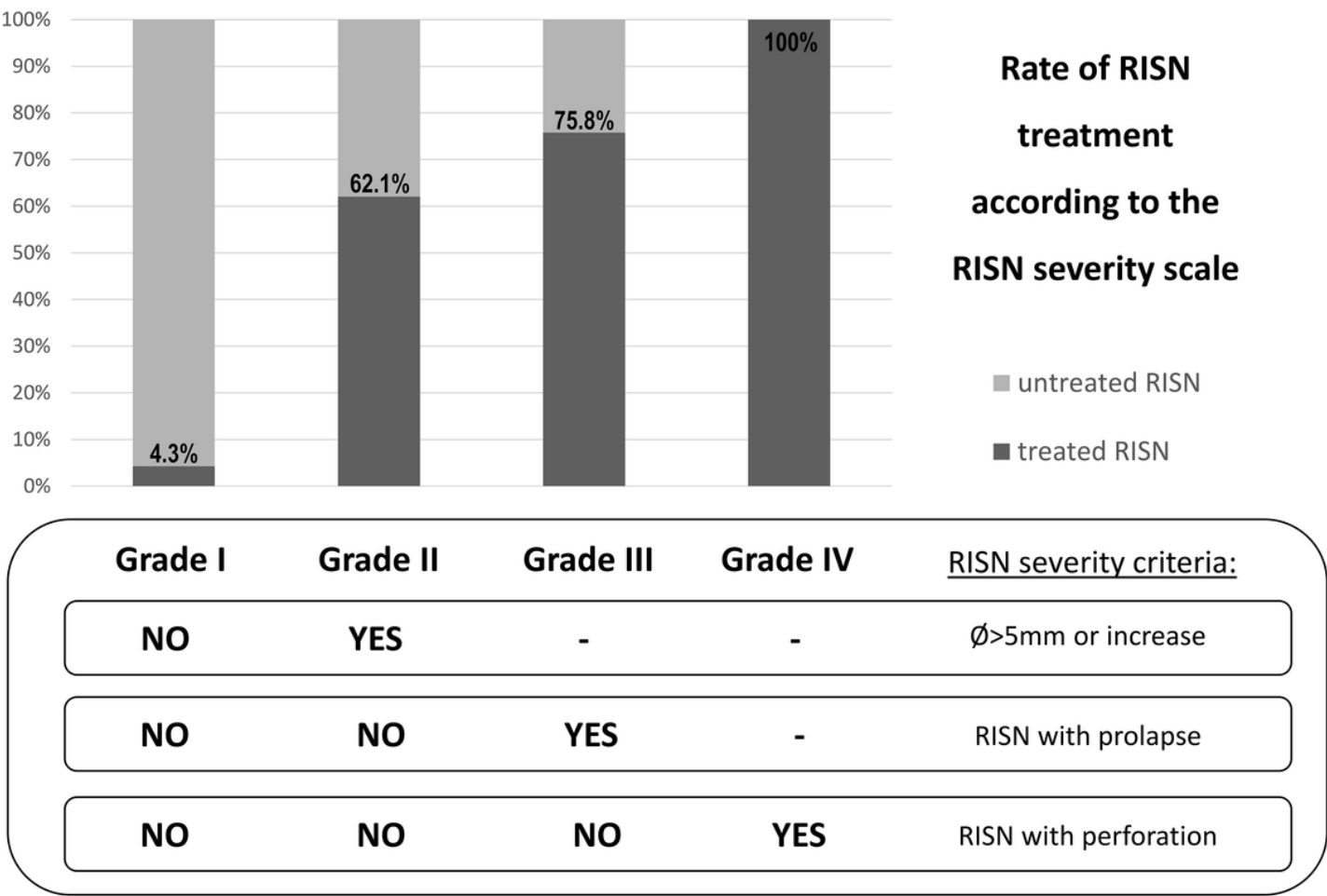


Figure 2

treatment rates of RISN in each severity grade, and presents a scheme for easy allocation of the RISN to an appropriate severity grade upon the relevant RISN characteristics (LD at the last follow-up, RISN size increase during the follow-up, presence of prolapse and globe perforation).

Abbreviations: RISN- radiation induced scleral necrosis; LD- largest dimension

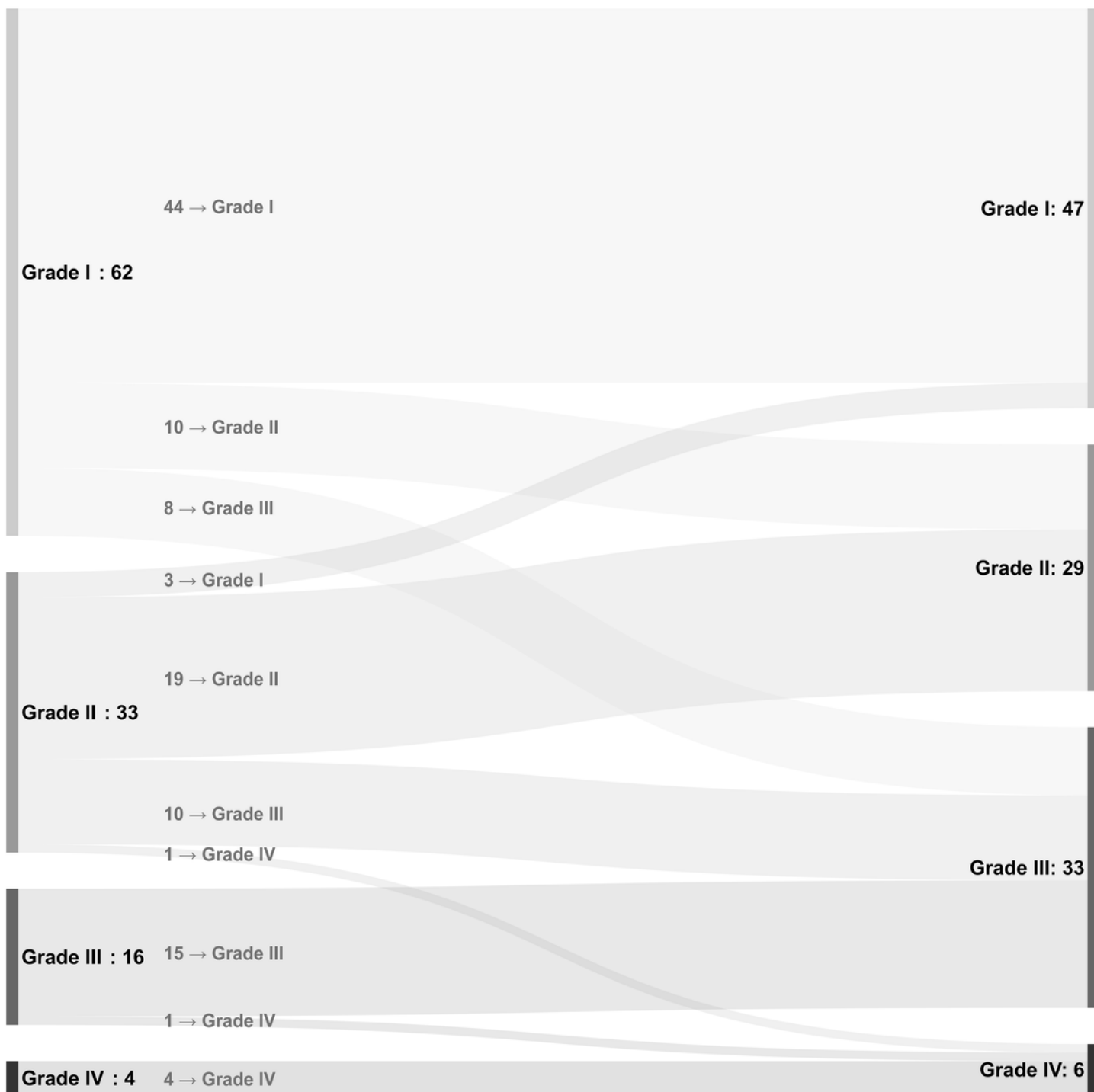


Figure 3

Sankey diagram showing the changes in the RISN severity scale in the analyzed cohort during the whole observational period.

Abbreviations: RISN- radiation induced scleral necrosis;

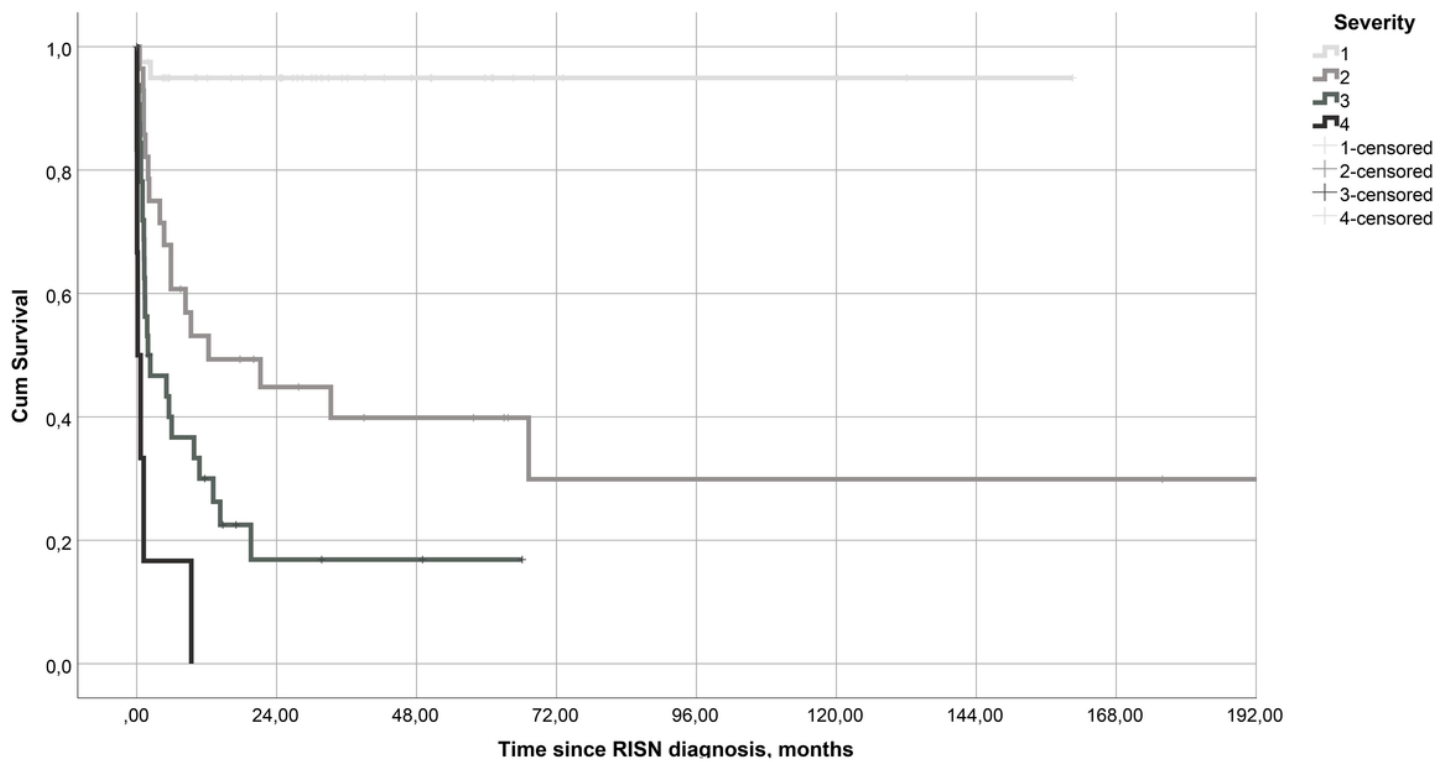


Figure 4

Kaplan-Meier survival plot showing the time course after RISN diagnosis until the surgical treatment in different subgroups depending on the values of the RISN severity scale (see the estimated survival times for each severity scale value in the supplementary table E1).

Abbreviations: RISN- radiation induced scleral necrosis

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